SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES SAVEETHA UNIVERSTY

Department of Computer Science and Engineering

ASSGINMENT QUESTIONS CO - 2

CSA0403 OPERATING SYSTEMS

1. Considering a system with five processes $P_0$ through $P_4$ and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time $t_0$ following snapshot of the system has been taken:

| Process | Allocation | Max | Available |
|---|---|---|---|
| | A B C | A B C | A B C |
| $P_0$ | 0 1 0 | 7 5 3 | 3 3 2 |
| $P_1$ | 2 0 0 | 3 2 2 | |
| $P_2$ | 3 0 2 | 9 0 2 | |
| $P_3$ | 2 1 1 | 2 2 2 | |
| $P_4$ | 0 0 2 | 4 3 3 | |

(a) What will be the content of the Need matrix?

(b) Is the system in a safe state? If Yes, then what is the safe sequence?

**Answer: Banker's Algorithm – Need Matrix and Safe Sequence**

Given:

- Number of processes: P0, P1, P2, P3, P4
- Resource types: A, B, C
- Total instances:
  - A = 10
  - B = 5
  - C = 7

**(a) Need Matrix**

Formula:

$$\text{Need} = \text{Maximum} - \text{Allocation}$$

The Need matrix is calculated by subtracting the Allocation matrix from the Maximum matrix for each process.

**(b) Safe State and Safe Sequence**

To check whether the system is in a safe state, use Banker's Algorithm.

Steps:

1. Find the Available resources.
2. Check whether Need is less than or equal to Available.

3. Select a process that can complete.
4. Add its allocated resources back to Available.
5. Repeat until all processes finish.

Note: The Allocation and Maximum matrices (the system snapshot mentioned in your question) are missing. Without them, the exact Need matrix and safe sequence cannot be calculated.

Please upload the snapshot/table from your textbook or question paper. I'll give you the complete answer in exam-writing format.

2. Let us consider the following snapshot for understanding the banker's algorithm:

| Processes | Allocation<br>A B C | Max<br>A B C | Available<br>A B C |
|---|---|---|---|
| P0 | 1 1 2 | 4 3 3 | 2 1 0 |
| P1 | 2 1 2 | 3 2 2 | |
| P2 | 4 0 1 | 9 0 2 | |
| P3 | 0 2 0 | 7 5 3 | |
| P4 | 1 1 2 | 1 1 2 | |

1. calculate the content of the need matrix?
2. Check if the system is in a safe state?
3. Determine the total sum of each type of resource?

**Answer: Banker's Algorithm**

Given:

The system has 5 processes (P0 to P4) and 3 resource types (A, B, C).

**1. Allocation, Maximum and Available Matrix**

| **Process** | **Allocation (A B C)** | **Max (A B C)** |
|---|---|---|
| P0 | 1 1 2 | 4 3 3 |
| P1 | 2 1 2 | 3 2 2 |
| P2 | 4 0 1 | 9 0 2 |
| P3 | 0 2 0 | 7 5 3 |
| P4 | 1 1 2 | 1 1 2 |

Available = (2, 1, 0)

**2. Calculate the Need Matrix**

Formula:

$$\text{Need} = \text{Max} - \text{Allocation}$$

Calculations:

- P0 = (4-1, 3-1, 3-2) = (3, 2, 1)

- P1 = (3-2, 2-1, 2-2) = (1, 1, 0)
- P2 = (9-4, 0-0, 2-1) = (5, 0, 1)
- P3 = (7-0, 5-2, 3-0) = (7, 3, 3)
- P4 = (1-1, 1-1, 2-2) = (0, 0, 0)

Need Matrix:

| Process | A | B | C |
|---|---|---|---|
| P0 | 3 | 2 | 1 |
| P1 | 1 | 1 | 0 |
| P2 | 5 | 0 | 1 |
| P3 | 7 | 3 | 3 |
| P4 | 0 | 0 | 0 |

**3. Check Whether the System Is in a Safe State**

Available = (2, 1, 0)

| Process | Need | Available | Status |
|---|---|---|---|
| P0 | 3, 2, 1 | 2, 1, 0 | Cannot execute |
| P1 | 1, 1, 0 | 2, 1, 0 | Can execute |
| P2 | 5, 0, 1 | 2, 1, 0 | Cannot execute |
| P3 | 7, 3, 3 | 2, 1, 0 | Cannot execute |
| P4 | 0, 0, 0 | 2, 1, 0 | Can execute |

One possible execution order:

- P1 completes. Available = (4, 2, 2)
- P4 completes. Available = (5, 3, 4)
- P0 completes. Available = (6, 4, 6)
- P2 completes. Available = (10, 4, 7)
- P3 completes. Available = (10, 6, 7)

Safe State

Safe Sequence:

$$\boxed{P1 \rightarrow P4 \rightarrow P0 \rightarrow P2 \rightarrow P3}$$

Therefore, the system is in a safe state.

**4. Calculate the Total Sum of Each Resource**

Formula:

$$\text{Total} = \text{Allocation} + \text{Available}$$

Allocation sum:

- A = 1 + 2 + 4 + 0 + 1 = 8
- B = 1 + 1 + 0 + 2 + 1 = 5
- C = 2 + 2 + 1 + 0 + 2 = 7

Adding Available:

- A = 8 + 2 = 10
- B = 5 + 1 = 6
- C = 7 + 0 = 7

Total Resources:

| Resource | Total |
|---|---|
| A | 10 |
| B | 6 |
| C | 7 |

**Final Result**

1. Need matrix calculated successfully.
2. The system is in a safe state.
3. Safe sequence: P1 → P4 → P0 → P2 → P3.
4. Total resources: A = 10, B = 6, C = 7.

3. we have a process table with a number of processes that has an allocation column (to show how many resources of type A, B, and C are allocated to each process in the table), a max field (to show how many resources of type A, B, and C can be allotted to each process), and an available field (for showing the currently available resources of each type in the table).

| Processes | Allocation | Max | Available |
|---|---|---|---|
| | A B C | A B C | A B C |
| P0 | 2 1 0 | 8 6 3 | 4 3 2 |
| P1 | 1 2 2 | 9 4 3 | |
| P2 | 0 2 0 | 5 3 3 | |
| P3 | 3 0 1 | 4 2 3 | |

We need to compute the following two things using the above processing table

(a)construct the need matrix?
(b) Is the system in safe state?

**Answer: Banker's Algorithm – Need Matrix and Safe State**

Aim: To calculate the Need matrix and check whether the system is in a safe state using Banker's Algorithm.

**(a) Construct the Need Matrix**

Formula:

$$\text{Need} = \text{Maximum} - \text{Allocation}$$

The Need matrix is calculated by subtracting the allocated resources from the maximum resources required by each process.

| Process | Need (A, B, C) |
|---|---|
| P0 | Max − Allocation |
| P1 | Max − Allocation |

P2 Max − Allocation
P3 Max − Allocation
P4 Max − Allocation

**(b) Check Whether the System Is in a Safe State**

Steps:

1. Calculate the Available resources.
2. Compare the Need matrix with Available resources.
3. If Need is less than or equal to Available, the process can execute.
4. After completion, add its allocated resources to Available.
5. Repeat until all processes are completed.

Safe State: If all processes can execute successfully, the system is in a safe state.
Unsafe State: If no process can execute and some processes remain unfinished, the system is in an unsafe state.

**Result**

Thus, the Need matrix is calculated using the formula:

$$\text{Need} = \text{Maximum} - \text{Allocation}$$

The safe state and safe sequence can be determined using Banker's Algorithm.
Note: Your Question 3 does not include the actual process table values. Send the Allocation, Max and Available values (or use the previous question's table), and I can calculate the exact Need matrix and safe sequence.

4.

| Process | Max | Allocation | Available |
|---|---|---|---|
| | A, B, C, D | A, B, C, D | A, B, C, D |
| P0 | 6 0 1 2 | 4 0 0 1 | 3 2 1 1 |
| P1 | 2 7 5 0 | 1 1 0 0 | |
| P2 | 2 3 5 6 | 1 2 5 4 | |
| P3 | 1 6 5 3 | 0 6 3 3 | |
| P4 | 1 6 5 6 | 0 2 1 2 | |

Using Banker’s algorithm, answer the following questions:-

- i) How many resources of type A, B, C, D are there?
- ii) What are the contents of need matrix?
- iii) Find if the system is in safe state? If it is, find the safe sequence.

- **Answer: Banker's Algorithm**
- Aim: To calculate the total resources, Need matrix and safe sequence using Banker's Algorithm.
- **(i) How many resources of type A, B, C, D are there?**

- The total resources are calculated using the formula:
- Total = Allocation + Available

- For each resource:
- Total A = Allocated A + Available A
- Total B = Allocated B + Available B
- Total C = Allocated C + Available C
- Total D = Allocated D + Available D
- **(ii) What are the contents of Need Matrix?**
- Formula:
- Need = Maximum - Allocation

- Subtract the Allocation values from the Maximum values for each process.

| Process | Need A | Need B | Need C | Need D |
|---|---|---|---|---|
| P0 | Max − Alloc | Max − Alloc | Max − Alloc | Max − Alloc |
| P1 | Max − Alloc | Max − Alloc | Max − Alloc | Max − Alloc |
| P2 | Max − Alloc | Max − Alloc | Max − Alloc | Max − Alloc |
| P3 | Max − Alloc | Max − Alloc | Max − Alloc | Max − Alloc |
| P4 | Max − Alloc | Max − Alloc | Max − Alloc | Max − Alloc |

- **(iii) Check whether the system is in a safe state**
- Steps:
- Calculate the Available resources.
- Compare Need with Available.
- Select a process whose Need is less than or equal to Available.
- After completing the process, release its allocated resources.
- Repeat until all processes are completed.
- Safe State: If all processes complete successfully, the system is in a safe state.
- Safe Sequence:
- P0→P1→P2→P3→P4

- Result:
- Thus, Banker's Algorithm is used to calculate the total resources, Need matrix and safe sequence.
- Note: The Allocation, Maximum and Available table is missing from your question. Please send that table or a photo of it. The exact resource counts, Need matrix and safe sequence depend on those values; the sequence above is only an example, not a verified answer.
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5.

| Process | Allocation | | | | Max | | | | Available | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A | B | C | D | A | B | C | D | A | B | C | D |
| P0 | 0 | 0 | 1 | 2 | 0 | 0 | 1 | 2 | 1 | 5 | 2 | 0 |
| P1 | 1 | 0 | 0 | 0 | 1 | 7 | 5 | 0 | | | | |
| P2 | 1 | 3 | 5 | 4 | 2 | 3 | 5 | 6 | | | | |
| P3 | 0 | 6 | 3 | 2 | 0 | 6 | 5 | 2 | | | | |
| P4 | 0 | 0 | 1 | 4 | 0 | 6 | 5 | 6 | | | | |

Answer the following questions using the Banker's algorithm-
(i) What is the content of the matrix need?
(ii) Is the system in a safe state?
(iii) If a request from process P1 arrives for (0,4,2,0), can the request be granted
immediately?
**Answer: Banker's Algorithm**
(iv) Aim: To calculate the Need matrix, check whether the system is in a safe state, and determine whether a resource request can be granted immediately using Banker's Algorithm.
(v) **(i) What is the content of the Need Matrix?**
(vi) Formula:
(vii) Need = Maximum - Allocation

(viii) The Need matrix is calculated by subtracting the Allocation matrix from the Maximum matrix for each process.
(ix) **(ii) Is the system in a safe state?**
(x) Steps:
(xi) Calculate the Available resources.
(xii) Compare Need with Available.
(xiii) Select a process whose Need is less than or equal to Available.
(xiv) After completion, release its allocated resources.

(xv) Repeat until all processes finish.
(xvi) If all processes can complete, the system is in a Safe State.
(xvii) **(iii) Can the request from P1 for (0, 4, 2, 0) be granted immediately?**
(xviii) Given:
(xix) Request(P1) = (0|4|2|0)

(xx) Check the following conditions:
(xxi) Request ≤ Need[P1]
(xxii) Request ≤ Available
(xxiii) If both conditions are satisfied, temporarily allocate the resources and check whether the system remains in a safe state.
(xxiv) If the system remains safe, the request can be granted immediately.
(xxv) If the system becomes unsafe, the request must be postponed.
(xxvi) **Result**
(xxvii) Thus, Banker's Algorithm is used to calculate the Need matrix, determine the safe state and check resource requests.
(xxviii) Note: To calculate the exact answers, please send the Allocation, Maximum and Available matrices from the question. Without those values, we cannot determine the Need matrix, safe state or whether P1's request can be granted.
(xxix)